SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

STUDENT DETAILS	
Name: MANDAPAKA SRI RAM (M. Sriram)	Reg No: 192311120
Course: Internet Programming (CSA43)	College: SIMATS Engineering

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
```

```
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

I **M.SriRam / 192311120** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ***M.SriRam (192311120)***

Name: MANDAPAKA SRI RAM Reg No: 192311120

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ASSESSMENT TOOL 3 - ANSWERS

Name: MANDAPAKA SRI RAM **Reg No:** 192311120 **Course:** Internet Programming (CSA43)

Question 1: Correct HTML tag for video embedding

Answer: Option B - <video src="video.mp4" controls></video>

Explanation

The <video> element was introduced in HTML5 to play video directly inside a web page without any third-party plugin such as Flash. The start-up wants to show a promotional video on the homepage, so the native <video> element is the standard, supported and future-proof choice. The **src** attribute points to the video file (video.mp4) and the **controls** attribute makes the browser show its built-in play/pause button, progress bar, volume and full-screen options.

Important attributes of the <video> tag

Attribute	Purpose
src	URL of the video file to play.
controls	Displays the browser's default playback controls.
autoplay / muted	Starts playing automatically (browsers normally allow this only when muted).
loop	Replays the video again when it ends.
poster	Image shown before the video starts or while it loads.
width / height	Reserves the display size and prevents layout shifts.
preload	Hints how much of the video to load in advance (none, metadata, auto).

Recommended code for the homepage

```
<video width="640" height="360" controls poster="promo.jpg">
  <source src="video.mp4" type="video/mp4">
  <source src="video.webm" type="video/webm">
  Your browser does not support the video tag.
</video>
```

Why the other options are wrong

Option	Verdict
A. <media>	Not an HTML element. The browser treats it as an unknown tag, so nothing is played.
C. <movie>	Not defined in any HTML standard; it is ignored by browsers.
D. <vid>	An invented tag; it has no meaning in HTML and does not display a video.

Conclusion

Only <video> is a valid, standards-compliant element, so **Option B** is correct. The text between the opening and closing tags works as a fallback message for very old browsers, and the controls attribute keeps the player accessible for users.

Question 2: Appropriate HTTP method for form submission

Answer: Option B - POST

Explanation

A registration form collects sensitive details such as name, e-mail and password, and these must be sent to the server securely. The **POST** method places the form data inside the **body** of the HTTP request instead of the URL. Because of this, the data is not shown in the address bar, is not stored in browser history or bookmarks and is not normally recorded in server access logs. POST is also intended for actions that **create** a new resource on the server, such as registering a new user.

GET versus POST

Feature	GET	POST
Data location	Appended to the URL as a query string	Sent in the request body
Visibility	Visible in URL, history and logs	Not visible in the URL
Size limit	Limited by URL length	No practical limit
Purpose	Retrieve / read data	Submit / create data
Bookmarkable	Yes	No
Suitable for passwords	No	Yes (with HTTPS)

Why PUT and DELETE are not suitable

- **PUT** replaces or updates an existing resource at a known URL; it is not used for ordinary HTML form submission (HTML forms only support GET and POST).
- **DELETE** removes a resource from the server, which is the opposite of registering a user.

Example code

```
<form action="/register" method="post">
  <label>Name:</label>  <input type="text" name="username" required>
  <label>Email:</label> <input type="email" name="email" required>
  <label>Password:</label> <input type="password" name="pwd" required>
  <button type="submit">Register</button>
</form>
```

What the browser sends

```
POST /register HTTP/1.1
Host: www.example.com
Content-Type: application/x-www-form-urlencoded

username=Sriram&email=sriram@example.com&pwd=*******
```

Conclusion

POST is the correct method because it keeps user data out of the URL and is designed for creating records. For real security the site should also use **HTTPS**, so that the request body is encrypted while travelling to the server.

Question 3: Proper semantic element for navigation

Answer: Option B - <nav>

Explanation

Semantic HTML means using elements whose names describe the meaning of their content. The <nav> element, introduced in HTML5, represents a section of a page that contains major navigation links such as the main menu, a table of contents or pagination. Using it tells the browser, search engines and assistive technologies that this block is a navigation menu, which is exactly what the start-up's requirement of a semantic navigation menu asks for.

Benefits of using <nav>

- **Accessibility:** screen readers announce a navigation landmark and let users jump directly to the menu.
- **SEO:** search engines understand the site structure and the importance of the links.
- **Readability and maintenance:** developers can understand the page structure at a glance, unlike a series of anonymous blocks.
- **Standards compliance:** follows the HTML5 specification and good web-development practice.

Example code

```
<nav aria-label="Main menu">
  <ul>
    <li><a href="index.html">Home</a></li>
    <li><a href="about.html">About</a></li>
    <li><a href="services.html">Services</a></li>
    <li><a href="contact.html">Contact</a></li>
  </ul>
</nav>
```

Why the other options are not the best choice

Option	Verdict
A.	A generic container with no meaning. It can hold a menu but gives no semantic information.
C. <section>	Groups related content under a heading (like a chapter); it is not specifically a navigation block.
D. <header>	Holds introductory content of a page or section (logo, title). A <nav> can be placed inside it, but the header itself is not the navigation element.

Conclusion

<nav> is the only element whose purpose is to define navigation links, so **Option B** is correct. Wrapping the links in a list (<ul> and <li>) inside the <nav> element and styling them with CSS gives a clean, accessible and standards-compliant menu.

Question 4: Correct XML syntax and errors in other options

Answer: Option B - <note><to>User</to><from>Admin</from></note>

Explanation

XML configuration files must be **well-formed**, otherwise an XML parser rejects the whole file. Option B follows every rule: the root element <note> is opened and closed, the child elements <to> and <from> each have matching start and end tags and the elements are correctly nested.

Rules of well-formed XML

- Every XML document must have exactly one root element.
- Every opening tag must have a matching closing tag.
- Tags must be properly nested (the last tag opened is the first one closed).
- Tag names are case-sensitive, so <To> and <to> are different.
- Attribute values must be placed inside quotes, and tags must be completed with >.

Error analysis of each option

Option	Code	Result / error
A	<note><to>User</to><from>Admin</from>	**Incorrect.** The root element <note> is never closed; the closing tag </note> is missing.
B	<note><to>User</to><from>Admin</from></note>	**Correct.** Proper nesting, all tags closed, single root.
C	<note><to>User</from></note>	**Incorrect.** <to> is closed by </from>, so the start and end tags do not match and nesting is broken.
D	<note to="User"></note	**Incorrect.** The closing tag is incomplete: </note has no closing > bracket.

Corrected and complete XML example

```
<?xml version="1.0" encoding="UTF-8"?>
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

Conclusion

Only **Option B** is a well-formed XML document. Fixing missing, mismatched or incomplete tags removes the syntax errors that the team found in the configuration files, and the file can then be parsed by any XML parser.

Question 5: Correct protocol for web communication

Answer: Option B - HTTP

Explanation

HTTP (HyperText Transfer Protocol) is the application-layer protocol that web browsers and web servers use to exchange web pages, images, videos and other content. A browser sends an HTTP **request** (for example GET or POST) and the server returns an HTTP **response** containing a status code and the requested content. It normally runs over TCP on port 80, and its secure version **HTTPS** uses TLS encryption on port 443. HTTP is stateless, so each request is independent, and cookies or sessions are used to remember users.

Simple request and response

```
GET /index.html HTTP/1.1
Host: www.example.com

HTTP/1.1 200 OK
Content-Type: text/html

<html> ... </html>
```

Why the other protocols are not suitable

Option	Protocol	Used for
A	FTP (File Transfer Protocol)	Uploading and downloading files between computers; not for delivering web pages.
C	SMTP (Simple Mail Transfer Protocol)	Sending e-mail between mail servers.
D	SNMP (Simple Network Management Protocol)	Monitoring and managing network devices such as routers and switches.

Conclusion

The application must deliver web content to browsers, so **HTTP** (preferably HTTPS) is the correct protocol. **Option B** is the answer.

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